

# MAT1190 Lecture Notes

Arky!! :3c

'26 Fall Semester

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## Day 1: Introduction (Sep. 8, 2026)

This is the first part of a two-semester course. Throughout this class, we will cover Hartshorne Chapter II. To quote Hunter Spink, “if I just went through Hartshorne straight, it would not make any sense; this textbook does not make any sense. As a first exposure, the book is not reasonable. ” We will get through scheme theory, though. The goal of this class is to develop scheme theory as a main topic, then show how it applies to a bunch of easier theories. “A lot of people think schemes are so confusing, that you basically have to do 6 to 9 on a completely obsolete theory to show you how it actually matters” (but he won’t do that).

Three references that are very useful, outside of Hartshorne.

- (i) Ravi Vakil’s *The Rising Sea*. This is meant to be a yearlong textbook.
- (ii) Jo Harris’ *The Geometry of Schemes*. Harris just says “this is schemes, suck it up” then develops a bunch of properties about it.
- (iii) Qin Li’s *Arithmetic* (something something).

The class will have an *oral* midterm and final (roughly October and December). There will be ungraded exercises (biweekly, as we go?) to turn in throughout the semester; the orals will be based on the easier questions.

This class will start with the theory of manifolds over  $\mathbb{R}$ , i.e., the classical theory of nice spaces. Consider the following insight on the Gelfand–Kolmogorov theorem: consider  $C^0(M, \mathbb{R})$ , the ring of continuous functions  $M \rightarrow \mathbb{R}$  on a manifold  $M$ . How do we recover  $M$  from  $C^0(M, \mathbb{R})$ ? Colloquially, different points of  $M$  see different aspects of any function in  $C^0(M, \mathbb{R})$ . If we start with a point  $p \in M$ , we get an  $\mathbb{R}$ -algebra map  $C^0(M, \mathbb{R}) \rightarrow \mathbb{R}$  given by evaluating the functions at the given point  $p$ ; we call this the “evaluation map”, denoted

$$\text{ev}_p: C^0(M, \mathbb{R}) \rightarrow \mathbb{R}, \quad f \mapsto f(p).$$

Specifically,  $\text{ev}_p \in \text{Hom}_{\mathbb{R}\text{-alg}}(C^0(M, \mathbb{R}), \mathbb{R})$ . What does it mean for  $\text{ev}_p$  to be an  $\mathbb{R}$ -algebra map? As an example, while  $\text{Hom}(\mathbb{C}, \mathbb{C}) = \text{Aut}(\mathbb{C})$  contains “uncountably many horrible things”, we see that  $\text{Hom}_{\mathbb{R}\text{-alg}}(\mathbb{C}, \mathbb{C})$  consists of homomorphisms that fix  $\mathbb{R}$ , i.e., it only admits the identity and complex conjugation.

**Claim 1.1.**  $\{\text{ev}_p\}_{p \in M}$  are the only elements of  $\text{Hom}_{\mathbb{R}\text{-alg}}(C^0(M, \mathbb{R}), \mathbb{R})$ .

We will do the proof for  $M = \mathbb{R}$ , then see that the proof follows analogously for several other examples of  $M$  as well.

*Proof.* Assume we have an  $\mathbb{R}$ -algebra map  $\Phi: C^0(M, \mathbb{R}) \rightarrow \mathbb{R}$  that sends the identity function  $x$  to  $a$ , i.e.,  $x \mapsto a$ . We claim that this map is  $\text{ev}_a$ ; if not, then there is some function  $g \in C^0(M, \mathbb{R})$  such that  $g \not\mapsto g(a)$  under  $\Phi$ . Indeed, consider  $\ker(\Phi)$ ; we see that  $x - a \in \ker(\Phi)$ ; now, consider the function

$$(x - a)^2 + (g(x) - b)^2,$$

where  $b := \Phi(g(x))$ . This function was constructed to never vanish; the only way for it to be zero would be for  $x = a$  and  $g(x) = \Phi(g(x))$ , but this never happens, so it is nonzero everywhere. However,  $x - a \in \ker \Phi$  and  $g(x) - b \in \ker \Phi$ , so

$$(x - a)^2 + (g(x) - b)^2 \in \ker \Phi,$$

whence it is a unit; this would mean

$$1 = \frac{1}{(x - a)^2 + (g(x) - b)^2} \cdot ((x - a)^2 + (g(x) - b)^2) \in \ker \Phi,$$

which is contradictory. □

Following Grothendieck, let  $R$  be any ring, and consider  $\text{Spec } R$ , which is a set of points with a topology, and is a locally ringed space (i.e., local functions “glue” to form global functions). We will rigorously define  $\text{Spec } R$  later. The key insight about manifolds is that, since they consist of charts that are compatible to each other locally, it allows us to “glue” data from functions together. Another important thing to know about locally ringed spaces is that if a function is non-vanishing everywhere, then it is invertible. Together, this lets you recover a topology on your ring  $R$ . **Note that this stuff is still intentionally hand-wavey for now.**<sup>1</sup>

Consider the polynomials over the real numbers,  $\mathbb{R}[x]$ . It is a ring. What would happen if we tried to reconstruct the space according to the procedure described earlier? Let’s look at the  $\mathbb{R}$ -algebra homomorphisms from  $\mathbb{R}[x] \rightarrow \mathbb{R}$ ; then we get that it is  $\mathbb{R}$ , since  $x \mapsto a \in \mathbb{R}$ , and there is probably some way to proceed from here.

There is, however, a reason we don’t do algebraic geometry over the real numbers; that reason being, the process would suggest that  $\mathbb{R}[x]$  is that ring of functions on the real line. If we looked at  $x^2 + 1 \in \mathbb{R}[x]$ , we would see that it’s a polynomial that doesn’t vanish anywhere and not invertible. This makes these “space-recovery” arguments impossible, specifically the part where a “non-vanishing function is invertible”. There are two solutions to this behavior... either you invert functions which don’t vanish everywhere, so you map  $\mathbb{R} \rightarrow \mathbb{R} \setminus \{0\}$ , or you can add points for every possible evaluation to every field. For example, consider

$$\text{ev}_i: \mathbb{R}[x] \rightarrow \mathbb{C}, \quad x \mapsto i;$$

we would see that  $x^2 + 1 \mapsto 0$ . Next time, we will define that  $\text{Spec}(R)$  is built out of all homomorphisms from  $\mathbb{R}$  to fields, modded out by “some stuff”.

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<sup>1</sup>“you... uncover something so unbelievably terrible that i can’t even begin to describe it. something that’s going to ruin your life for the next... twelve weeks!”

## Day 2: Spectrum of a Ring (Sep. 10, 2026)

Last class, we talked about evaluation maps. Let  $M$  be a manifold, and let  $p \in M$ . We have that all of the  $\mathbb{R}$ -algebra maps  $C^0(M, \mathbb{R}) \rightarrow \mathbb{R}$  were given by  $\text{ev}_p: C^0(M, \mathbb{R}) \rightarrow \mathbb{R}$ . A key fact about locally ringed spaces is that any function on them which is nonvanishing everywhere is invertible; when working with polynomials  $\mathbb{R}[x]$ , though, we see that  $x^2 + 1$  constitutes a counterexample to the aforementioned statement.

The function  $x^2 + 1$  maps  $\mathbb{R}$  to  $\mathbb{R} \setminus \{0\}$ . Observe that we may identify  $xy - 1 = 0$  with  $\mathbb{R} \setminus \{0\}$  by the isomorphism  $(x, y) \mapsto x$ ; the curve  $xy - 1 = 0$  (visualized, perhaps, in the  $xy$ -plane) admits an automorphism  $(x, y) \mapsto (y, x)$ ; if we pre-compose  $(x, y) \mapsto x$  with this, we see that we get an automorphism on  $\mathbb{R} \setminus \{0\}$  in the form of  $x \mapsto x^{-1}$ . This is called the *Rabinowitsch trick*.

Our solution to the above problem is to “add more points”. Consider  $\text{Hom}_{\mathbb{R}\text{-alg}}(\mathbb{R}[x], \mathbb{C})$  (note that we extended the codomain  $\mathbb{R}$  to  $\mathbb{C}$ ). Fix a field  $K$ , and consider  $f_1, \dots, f_r \in K[x_1, \dots, x_n]$ , and consider the set

$$\{f_1(x_1, \dots, x_n) = f_r(x_1, \dots, x_n) = 0\} \subset K^n.$$

This is the zero set  $Z_K(f_1, \dots, f_r)$ .

**Theorem 2.1.** We have the canonical bijection,

$$\text{Hom}_{K\text{-alg}}\left(\frac{K[x_1, \dots, x_n]}{(f_1, \dots, f_r)}, K\right) \cong Z_K(f_1, \dots, f_r).$$

*Proof.* The key fact about polynomial rings is that, in general, for any ring  $R$ ,  $\text{Hom}_{R\text{-alg}}(R[x_1, \dots, x_n], S)$  is in a very natural bijection with  $n$ -tuples of elements of  $S$ . Concretely, you can send  $(x_1, \dots, x_n) \mapsto (s_1, \dots, s_n)$  because  $x_1, \dots, x_n$  don't have to satisfy any relation over  $R$ . In this manner, any map from the quotient ring to  $K$  (in the original theorem statement) must send  $f_1, \dots, f_r$  to 0. Per our key fact, it follows that  $f \mapsto \text{ev}_{(s_1, \dots, s_n)}(f) = f(s_1, \dots, s_n)$ .

The canonical bijection thus arises for  $(a_1, \dots, a_n) \in Z_K(f_1, \dots, f_r)$  and its associated evaluation maps  $\text{ev}_{(a_1, \dots, a_n)}$ .  $\square$

Thus, we want to consider  $K[x_1, \dots, x_n]/(f_1, \dots, f_r)$  to be the “ring of functions”. We want to ask, what is the function  $f_1$  on the space  $Z_K(f_1, \dots, f_r)$ ? Clearly,  $f_1|_{Z_K(f_1, \dots, f_r)} = 0$  (and the same holds for each of  $f_1, \dots, f_r$ ).

As an example, consider a field  $K$ ,  $f_1 = x$ , and  $f_2 = y$ . Then  $Z_K(f_1, f_2) = (0, 0)$ , and  $K[x, y]/(f_1, f_2) \cong K$ . If  $f_1 = x$  and  $f_2 = y^2$ , then

$$\frac{K[x, y]}{(x, y^2)} \cong \frac{K[y]}{y^2}.$$

Here,  $\text{ev}_{(0,0)}(y) = 0$ , but we aren't setting  $y$  to 0 (as we are quotienting out by  $y^2$ ). Thus, we have a hurdle to deal with; (not all) functions will not be determined by their points.

Let  $L/K$  be a field over  $K$ , and consider  $Z_L(f_1, \dots, f_r)$ , where  $f_1, \dots, f_r$  are polynomials over  $K$ . Since  $K$  is embedded in  $L$ , we can consider  $f_1, \dots, f_r$  as polynomials over  $L$  as well, and write

$$Z_L(f_1, \dots, f_r) = \{(a_1, \dots, a_n) \mid f_1(a_1, \dots, a_n) = \dots = f_r(a_1, \dots, a_n) = 0\} \subset L^n.$$

He then drew a diagram.

**Theorem 2.2.** (i)  $\text{Hom}_{K\text{-alg}}(K[x_1, \dots, x_n]/(f_1, \dots, f_r), L)$  is in bijection with  $Z_L(f_1, \dots, f_r)$ .

(ii) Let  $R = K[x_1, \dots, x_n]/(f_1, \dots, f_r)$ . We have that

$$\text{Hom}_{K\text{-alg}}(R, L) \cong \{(\text{prime ideals } p \subset R), \kappa(p) \xrightarrow{K\text{-alg}} L\},$$

where  $\kappa(p) = \text{Frac}(R/p)$ .

Basically, this theorem says that “instead of being really good at solving system of equations, you could be really good at finding prime ideals instead.”

*Proof.* For (i), the proof is the same as earlier. For (ii), we have a  $K$ -algebra map  $\Phi: R \rightarrow L$ ; because the image is a field, we can factor the map through intermediate rings canonically. Because  $\text{Im } \Phi$  is an integral domain,  $\Phi$  factors through

$$R \rightarrow R/p \xrightarrow{K\text{-alg}} L,$$

where  $\text{Im } \Phi = R/p$ . In particular, since  $L$  is a field, we can further write

$$R \rightarrow R/p \hookrightarrow \text{Frac}(R/p) \xrightarrow{K\text{-alg}} L.$$

The first two maps are canonical to the prime ideal  $p$ . □

“To solve a system of equations over a larger field, we specify a prime ideal, then specify an embedding.”

**Definition 2.3.** We define  $|\text{Spec}(R)| := \{p \subset R \mid p \text{ prime}\}$ .

**Corollary 2.4.** If  $R$  is a field, then  $|\text{Spec}(R)|$  is a single point.

Of all the points that constitute  $\text{Spec}(R)$ , of which they are prime ideals, some of them are  $\ker \text{ev}_p$  for  $p \in Z_K(f_1, \dots, f_r)$ .

As an example, consider  $|\text{Spec } \mathbb{R}[x]|$ .  $\mathbb{R}[x]$  is a UFD, so its prime ideals are of the form  $(f)$  with  $f$  irreducible. Specifically,  $|\text{Spec } \mathbb{R}[x]| = \{(x - a) \mid a \in \mathbb{R}\} \cup \{(x - a)^2 + b^2 \mid a \in \mathbb{R}, b \sim -b \in \mathbb{R}^*\}$  (i.e., we identify  $b$  with  $-b$  because it doesn't matter here). Following our process from earlier, we have

$$\mathbb{R}[x] \rightarrow \text{Frac}\left(\frac{\mathbb{R}[x]}{(x - a)}\right) \xrightarrow{\mathbb{R}\text{-alg}} L,$$

where  $L/\mathbb{R}$ . Here,  $\text{Frac}(\mathbb{R}[x]/(x - a)) \cong \mathbb{R}$ . For the irreducible quadratics, we have

$$\mathbb{R}[x] \rightarrow \text{Frac}\left(\frac{\mathbb{R}[x]}{((x - a)^2 + b^2)}\right) \xrightarrow{\mathbb{R}\text{-alg}} L,$$

where  $\text{Frac}(\mathbb{R}[x]/((x - a)^2 + b^2)) \cong \mathbb{C}$ . Thus,  $|\text{Spec } \mathbb{R}[x]|$  is given by  $\mathbb{C}$  modded out by complex conjugation, with  $\{0\}$  added in.

Now, consider

$$\mathbb{R}[x] \rightarrow \text{Frac}(\mathbb{R}[x]/(0)) \hookrightarrow L,$$

where  $X$  satisfies no algebraic equation over  $\mathbb{R}$ .

### Day 3: Zariski Topology (Sep. 15, 2026)

It turns out that Hunter Spink types out his lecture notes! Regardless, I'll try to keep up here (until I burn out).

Recall that we defined  $|\text{Spec } R|$  to be the collection of prime ideals in  $R$ . For example,  $|\text{Spec } \mathbb{C}[x]|$  is given by  $\mathbb{C}$  modded out by complex conjugation, along with the  $\{0\}$  ideal itself (we call this the transcendental solution to  $Z(\emptyset)$ ).

Recall that  $\text{Hom}_{\mathbb{R}\text{-alg}}(C^0(M, \mathbb{R}), \mathbb{R}) = \{\text{ev}_p : p \in M\}$ . The idea being, if you want to identify  $Z(f)$ , we can simply look at all the points  $p$  such that  $\text{ev}_p f = 0$  instead. Specifically, we want to look at the bijection between  $\{\text{ev}_p : p \in M\}$  and  $\{p \in M\}$ .

Consider  $|\text{Spec } K|$ , where  $K$  is a field. Clearly,  $|\text{Spec } K| = \{(0)\}$ . Using the fact that  $K(p) = \text{Frac}(R/p)$  and  $f \in R$ , we have that  $f(p)$  is the inverse of  $f$  in  $K(p)$ , i.e.,  $\text{ev}_p : R \rightarrow K(p)$ .

**Definition 3.1.** For  $\mathcal{A} \subset R$ , we define  $Z(\mathcal{A})$  to be

$$Z(\mathcal{A}) = \{p \in \text{Spec } R \mid f(p) = 0, \forall f \in \mathcal{A}\} \subset \text{Spec } R.$$

Specifically, the above definition is equivalent to  $p \in \text{Spec } R$  such that  $\mathcal{A} \subset p$ . Think of the above as “all functions that vanish at  $p$ ”.

We define the Zariski topology on  $|\text{Spec } R|$  to have  $Z(\mathcal{A})$  as the closed sets. This is a *tentative definition*, and will be expanded upon later. We want to show that this is actually a topology!

**Theorem 3.2.** The Zariski topology (described above) is a topology.

*Proof.* First, we want to show that  $\emptyset$  and  $|\text{Spec } R|$  lie in the Zariski topology. For  $\emptyset$ , we can take  $Z(R)$ , or really just  $Z(1)$  (1 being picked because it is a unit). 1 has the great feature that it never vanishes in any field. For  $|\text{Spec } R|$ , we can take  $Z(0) = Z(\emptyset)$ .

For finite unions (since we're prescribing closed, not open sets), we have that  $Z(\mathcal{A}) \cup Z(\mathcal{B}) = Z(\mathcal{A}\mathcal{B})$ . For arbitrary intersections, we have

$$\bigcap Z(\mathcal{A}_\beta) = Z\left(\bigcup \mathcal{A}_\beta\right).$$

Now, we want to check that

$$Z(\mathcal{A}\mathcal{B}) = \{p \mid fg \in p, \forall f \in \mathcal{A}, g \in \mathcal{B}\},$$

and

$$Z(\mathcal{A}) \cup Z(\mathcal{B}) = \{p \mid f \in p, \forall f \in \mathcal{A}\} \cup \{p \mid g \in p, \forall g \in \mathcal{B}\},$$

in order to check that our statement for finite unions holds; indeed, this follows from the fact that, if there exists  $f \in \mathcal{A}$  with  $f \notin p$  and  $g \in \mathcal{B}$  with  $g \notin p$ , then  $fg \notin p$ .  $\square$

We want to look at some examples of topological spaces now. As an example, consider  $\text{Spec } \mathbb{Z}$ . This space has one point for every prime ideal in  $\mathbb{Z}$ , of which are the zero ideal and ideals generated by primes. What does  $Z(4)$  look like here?  $\bar{4} = 0$  in  $(2) = \mathbb{F}_2$ ,  $\bar{4} = 1$  in  $(3) = \mathbb{F}_3$ , and so on, with it simply being 4 in  $(0) = \mathbb{Z}$ . Thus,  $Z(4) = \{(2)\}$  (i.e., check the prime factorization).<sup>2</sup> We can try this with other numbers too, such as  $Z(12) = \{(2), (3)\}$ ,  $Z(1) = \emptyset$ , and  $Z(0) = \text{Spec } \mathbb{Z}$ .

The closed sets here are  $\emptyset$ ,  $\text{Spec } \mathbb{Z}$ , and finite collections of primes. In particular,  $\overline{\{(0)\}} = \text{Spec } \mathbb{Z}$ . At this point, Hunter made a remark that the facts we know about manifold topology don't apply here at all, which makes sense, since this space is... really weird.

<sup>2</sup>four, four, four, four... (draws a lot of 4s on the blackboard). JHIN!!!!

Recall that

$$|\operatorname{Spec} \mathbb{R}[x]| = \left( \frac{\mathbb{C}}{z \leftrightarrow \bar{z}} \right) \cup \{(0)\}.$$

Here,  $Z(f) = Z(f_1) \cup \dots \cup Z(f_k)$ , given that we can factor  $f$  as  $f_1 f_2 \dots f_k$  with each  $f_i$  irreducible. For irreducible  $f$ , if  $f = x - a$ , then  $Z(f) = \{(x - a)\}$ ; if  $f = (x - a)^2 + b^2$ , then  $p = (f)$ . Both  $\mathbb{Z}$  and  $\mathbb{R}[x]$  are UFDs; in particular, they are Dedekind domains, i.e., all nonzero proper ideals can be factored into a product of prime ideals.

We're now going to do an example where the ring is not a UFD. Consider  $|\operatorname{Spec} \mathbb{C}[x, y]|$  (where  $\mathbb{C}[x, y]$  is a UFD); per the Nullstellensatz, there are three types of primes. We can think of the dimension 0, 1, and 2 varieties, which will correspond to these prime ideals. We classify by height of primes, i.e., the transcendence degree of  $\kappa(p)/\mathbb{C}$ . For dimension 0 varieties, we can pick out individual points  $(a, b)$  of the affine space  $\mathbb{A}^2$ . For  $f$  irreducible,  $\overline{\{(f)\}} = (f) \cup \{(x - a, y - b) \mid f(a, b) = 0\}$ . "As a topological space, this thing is a disaster."

When does  $Z(g)$  contain  $(f)$ ? This occurs when  $g = 0$  in  $\operatorname{Frac}(\mathbb{C}[x, y]/(f))$ , which is equivalent to  $g \in (f)$ . Then  $g$  is a multiple of  $f$ . Since  $\mathbb{C}[x, y]$  is a UFD, the smallest  $Z(s)$  contains  $f$  if  $Z(f)$  is given by  $p$  such that  $f(p) = 0$ , i.e.,  $\{(x - a, y - b), f(x, y) = 0, (f)\}$ . You can always find closures from first principles, but to be more efficient you need "non-tautologous math".

## Day 4: Spectrum Properties (Sep. 17, 2026)

Recall, from last time, that we took

$$|\mathrm{Spec} R| = \{p \in R \mid p \text{ is prime}\},$$

and gave it a topology with a basis of closed sets  $Z(f) = \{p \in \mathrm{Spec} R \mid f(p) = 0\}$ , where  $f \in R$ , and  $f(p) = 0$  meant the image of  $f$  in  $k(p) = \mathrm{Frac}(R/p)$ . Equivalently,  $f \in p$ .

**Theorem 4.1.** If  $I = (A)$  is the ideal generated by  $A \subset R$ , then  $z(A) = z(I) = \{p \in \mathrm{Spec} R \mid I \subset p\}$ .

*Proof.*  $z(A) = \{p \in \mathrm{Spec} R \mid A \subset p\} = \{p \in \mathrm{Spec} R \mid I \subset p\} = z(I)$ . □

**Theorem 4.2.** The assignment  $R \rightarrow |\mathrm{Spec} R|$  is a contravariant functor **Ring**  $\rightarrow$  **Top**.

We introduce some category theory here. A *category* is a set of objects  $\mathrm{Ob}(C)$ , a collection of morphisms  $\{\mathrm{Mor}(X, Y) : X, Y \in \mathrm{Ob}(C)\}$ , and a composition map

$$\circ : \mathrm{Mor}(X, Y) \times \mathrm{Mor}(Y, Z) \rightarrow \mathrm{Mor}(X, Z)$$

satisfying the properties,

- (i) Composition is associative, i.e.,  $f \circ (g \circ h) = (f \circ g) \circ h$  whenever this makes sense,
- (ii) An identity map exists, i.e.,  $\mathrm{id}_X \in \mathrm{Mor}(X, X)$  such that  $\mathrm{id}_X \circ f = f = f \circ \mathrm{id}_X$  whenever  $f \in \mathrm{Mor}(X, Y)$ .

In the theorem above, there are two categories: rings and topological spaces. **Ring** is a category whose objects are rings and morphisms are ring maps (there's nothing to check here because the morphisms are determined at the level of elements). In particular, **Ring** is a subcategory of **Set**. **Top** is a category whose objects are topological spaces and morphisms are continuous maps.

A *functor*  $F : \mathcal{C} \rightarrow \mathcal{D}$  between categories is the data of two maps (both denoted  $F$ )

$$F : \mathrm{Ob}(\mathcal{C}) \rightarrow \mathrm{Ob}(\mathcal{D}), \quad F : \mathrm{Mor}(X, Y) \rightarrow \mathrm{Mor}(F(X), F(Y)),$$

satisfying the following relations,

- (i)  $F(f) \circ F(g) = F(f \circ g)$  when this makes sense,
- (ii)  $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$ .

We now define the notion of an *opposite category*. Let  $\mathcal{D}$  be a category. Then  $\mathcal{D}^{\mathrm{op}}$  as a category consists has  $\mathrm{Ob}(\mathcal{D}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{D})$  and

$$\mathrm{Mor}_{\mathcal{D}^{\mathrm{op}}}(X, Y) := \{f^{\mathrm{op}} \mid f \in \mathrm{Mor}_{\mathcal{D}}(Y, X)\}.$$

Composition works by  $f^{\mathrm{op}} \circ g^{\mathrm{op}} := (g \circ f)^{\mathrm{op}}$ . With this, we may now define contravariant functors. A *contravariant functor* is a functor  $F : \mathcal{C} \rightarrow \mathcal{D}^{\mathrm{op}}$  with

$$F : \mathrm{Ob}(\mathcal{C}) \rightarrow \mathrm{Ob}(\mathcal{D}), \quad F : \mathrm{Mor}(X, Y) \rightarrow \mathrm{Mor}(F(Y), F(X)),$$

satisfying obvious rules  $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$  and  $F(f \circ g) = F(g) \circ F(f)$ . As an example of more functors, we have the forgetful functor **Ring**  $\rightarrow$  **Set**, which is basically a functor that “forgets the ring structure”. We also have the contravariant functor from **Vect**<sub>K</sub>  $\rightarrow$  **Vect**<sub>K</sub>, where we sent  $V \rightarrow V^* = \mathrm{Hom}(V, K)$ , given by the dual map  $(f : V \rightarrow W) \mapsto (f^* : W^* \rightarrow V^*)$ .

As another example, consider the functor  $\mathbf{Mfld} \rightarrow \mathbf{Ring}$ , given by  $F: M \rightarrow C^0(M, \mathbb{R})$ , and  $(\Phi: M \rightarrow N) \mapsto (\Phi^*: C^0(N, \mathbb{R}) \rightarrow C^0(M, \mathbb{R}))$ , i.e., the pullback.

We now prove 4.2.

*Proof.* Given a map  $f: R \rightarrow S$ , we can define  $f^*: \text{Spec } S \rightarrow \text{Spec } R$  by  $q \mapsto f^{-1}(q)$ . We want to claim that  $f^{-1}(q)$  is actually a prime ideal. Suppose  $ab \in f^{-1}(q)$ ; equivalently,  $f(ab) \in q$ ; since  $f$  is a ring map, this is equivalent to saying that  $f(a) \in q$  or  $f(b) \in q$ , so  $a \in f^{-1}(q)$  or  $b \in f^{-1}(q)$ . Thus,  $f^{-1}(q) \in \text{Spec}(R)$ .

We now want to check that  $f^*$  is a continuous map of topological spaces. Take a closed set  $z(I) \subset \text{Spec}(R)$ ; we want to look at  $(f^*)^{-1}(z(I))$ . By definition,

$$(f^*)^{-1}(z(I)) = \{q \in \text{Spec } S \mid f^*(q) \in z(I)\}.$$

$f^*(q)$  is  $f^{-1}(q)$ , so  $f^*(q) \in z(I)$  is equivalent to  $I \subset f^{-1}(q)$ , which is the same as  $f(I) \subset q$ . Thus, the above set is simply  $z(f(I))$ .

Finally, we want to check that we actually have a contravariant functor. This is really easy, since the map we have is defined by set maps. First,  $(\text{id}_R)^* = \text{id}_{\text{Spec } R}$ ; second, if we have a map  $g: R \rightarrow S$  and  $f: S \rightarrow T$ , then  $f^*: \text{Spec } T \rightarrow \text{Spec } S$  and  $g^*: \text{Spec } S \rightarrow \text{Spec } R$ , and we have  $(f \circ g)^* = g^* \circ f^*$  as desired. Indeed, for a prime ideal  $t \subset T$ , we have  $(f \circ g)^*(t) = (f \circ g)^{-1}(t) = g^{-1} \circ f^{-1}(t) = g^* \circ f^*(t)$ .  $\square$

“Yay, we’ve created half the definition for an affine scheme so far.”

**Theorem 4.3.** Suppose  $\Phi: R \rightarrow S$ . Then for  $f \in R$ , we have  $\Phi(f) = f_{|\text{Spec } R|} \circ \Phi^*$ , where we consider  $f: \text{Spec } R \rightarrow \prod_{p \in \text{Spec}(R)} k(p)$ .

$$\begin{array}{ccc} S & \xleftarrow{\Phi(f)} & R \\ \vdots & & \vdots \\ \text{Spec } S & \xrightarrow{\Phi^*} & \text{Spec } R \end{array}$$

*Proof.* Hunter said he would leave this as an exercise once he figures out the statement (why...).  $\square$

**Theorem 4.4.** Let  $Y \subset \text{Spec } R$ . Then  $\overline{Y} = Z\left(\bigcap_{p \in Y} p\right)$ ,<sup>3</sup>

*Proof.*  $\overline{Y} = z(I)$ . Suppose  $z(J) \supset Y$ ; equivalently,  $J$  lies in each prime ideal of  $Y$ , which we can write as

$$J \subset \bigcap_{p \in Y} p. \quad \square$$

**Corollary 4.5.**  $\overline{\{p\}} = z(p) = \{q \in \text{Spec } R \mid p \subset q\}$ .

Intuitively, “ $p$  is a generic point inside its closure  $\overline{\{p\}}$ ”. As an example from last class, if we look at  $\text{Spec } \mathbb{C}[x, y]$ , then  $\overline{\{f\}}$  is given by the points on the curve  $\bigcup \{(f)\}$ . He then said something about the map  $\mathbb{C}[x, y]/(f) \rightarrow \text{Frac}(\mathbb{C}[x, y]/(f))$ , but I didn’t catch it.

**Corollary 4.6.** If  $\overline{z} \subset z(p)$ , then every open set containing  $\overline{z}$  contains  $p$ .

As a corollary to our discussion, a prime ideal  $p$  is closed if and only if  $p$  is maximal.

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<sup>3</sup>i swear to god if i should’ve been using  $\mathfrak{p}$  instead of  $p$  this whole time i will cry

Let's look at an example of an interesting ring that'll we'll see a lot in the future. Recall that if  $K$  is a field, then  $\text{Spec } K$  is a single point  $\{(0)\}$ . The next most complicated space, as spectrums go, is a local ring.

**Definition 4.7.** A *local ring* is a ring with exactly one maximal ideal.

**Corollary 4.8.**  $\text{Spec } R$  has a unique closed point if  $R$  is a local ring.

As an example,  $K[X]_{(x-5)} = \{\frac{f}{g} : f, g \in K[X], (x-5) \text{ divides } g \text{ as many times as it divides } f\}$ .  $(x-5) \subset K[X]_{(x-5)}$  is a maximal ideal because the kernel of the evaluation map  $\text{ev}_5 : K[X]_{(x-5)} \rightarrow K$  is  $(x-5)$ . Note that, if  $f(5) \neq 0$ , then  $f$  is a unit. Now,

$$\frac{f}{g} = (x-5)^a \frac{\tilde{f}}{\tilde{g}},$$

where  $\tilde{f}$  and  $\tilde{g}$  don't vanish at 5. Indeed,  $\frac{1}{\tilde{f}} \in K[X]_{(x-5)}$ .

Given any ideal  $I \subset K[X]_{(x-5)}$  generated by  $\mathcal{A}$ , replace each element of  $\mathcal{A}$  with a power of  $x-5$ ; then  $I$  is generated by some list of powers of  $x-5$ . This means  $I = (x-5)^a$ , or  $I = (a)$ , meaning if  $a \geq 2$ , then it is not prime, since  $(x-5)(x-5)^{a-1} \in ((x-5)^a)$ . In this manner, in

$$\text{Spec } K[X]_{(x-5)} = \{(0), (x-5)\},$$

the closed sets are  $(0)$  (i.e., the whole space),  $(x-5)$ , and  $\emptyset$ .

We then gave a small preview of the structure sheaf. We now end with a final "easy observation".

**Theorem 4.9.**  $\text{Spec } R/I$  embeds into  $\text{Spec } R$  with the subspace topology as  $z(I)$ .

"Next week, we're going to do the real meat and potatoes of... spectrums."

## Day 5: Localization (Sep. 22, 2026)

Given a function  $\phi: R \rightarrow S$  (thought of as function pullback), we have that  $\phi^*: |\text{Spec } S| \rightarrow |\text{Spec } R|$  and  $(\phi^*)^{-1}Z(I) = Z(\phi(I))$ .

**Definition 5.1.** Given  $f \in R$ , define the basic open subset

$$D(f) := Z(f)^c = \{\mathfrak{p} \in \text{Spec } R \mid f(\mathfrak{p}) \neq 0\}.$$

The  $D(f)$ 's form a subset of the topology; i.e., every open set is a union of these  $D(f)$ , and we have

$$D(f_1) \cap D(f_2) \cap \cdots \cap D(f_k) = D(f_1 \cdots f_k).$$

**Theorem 5.2** (Rabinowitsch). Let  $f_1, \dots, f_n, g \in k[x_1, \dots, x_n]$ ; then

$$Z_k(f_1, \dots, f_n) \cap D_k(g) = Z_k(f_1, \dots, f_n, x_{n+1}g - 1).$$

Specifically, the left hand side is a subset of  $k^n$ , and the right hand side is a subset of  $k^{n+1}$ .

*Proof.* The proof is simple; consider the correspondence given by the two maps

$$(a_1, \dots, a_n) \mapsto \left(a_1, \dots, a_n, \frac{1}{g(a_1, \dots, a_n)}\right), \quad (b_1, \dots, b_n) \mapsto (b_1, \dots, b_{n+1}). \quad \square$$

**Definition 5.3.** We define the localization of  $R$  at  $g$  as  $R_g := R[x]/(xg - 1)$ . We will denote  $x$  as  $g^{-1}$  freely.

**Theorem 5.4.** We have the following properties for localization,

- (i) Every element of  $R_g$  can be written as  $rg^{-1}$  with  $r \in R$ .
- (ii)  $rg^{-1} \in Sg^{-1}$  if and only if  $g^N(r - s) = 0$  for some  $N$ .
- (iii) A ring map  $R_g \rightarrow S$  is the same as a map  $R \rightarrow S$  such that  $g$  maps to a unit. Specifically, this is a universal property

$$\begin{array}{ccc} R & \xrightarrow{\quad} & R_g \\ & \searrow \text{sends } g \text{ to unit} & \downarrow \exists! \\ & & S \end{array}$$

- (iv) This map from  $R \rightarrow R_g$  is an isomorphism if and only if  $g$  is a unit. In words, “if you tried to localize at a unit, you didn’t actually do anything”.

*Proof.* For (i), take some element of the localization  $R_g$ , and write it as  $f(x) = r_0 + r_1x + \cdots + r_nx^n$ . We can factor out  $x^n$  by writing

$$f(x) = (r_0g^n + \cdots + r_1g^{n-1} + \cdots + r_n)x^n,$$

for which the bracketed portion is in  $R$ , and  $x^n = g^{-n}$ , so we are done.

For (ii), by taking the difference of two sides, it suffices to show that  $r = 0$  in  $R_g$  if and only if  $g^N r = 0$  in  $R$ , or  $r = 0$  in  $R_g$ . We really just want to look at the kernel of the map from  $R \rightarrow R_g$ , consisting of the elements which are annihilated by some power of  $g$ . If  $r \mapsto 0$  in  $R[x]/(xg - 1)$ , then in  $R[x]$ ,  $r = (1 - xy)(a_0 + a_1x + \cdots + a_nx^n)$ ; by comparing coefficients, we get

$$r = a_0 + (a_1 - ga_0)x + (a_2 - ga_1)x^2 + \cdots + (a_n - ga_{n-1})x^n - a_nx^{n+1}.$$

This tells us that  $a_0 = r$ ,  $a_1 = gr$ ,  $a_2 = g^2r$ , etc., so  $a_n = g^n r$ , which we know is 0. The converse is obvious: if  $g^N r = 0$  in  $R$ , then  $g^N r = 0$  in  $R_g$ , i.e.,  $r = 0$  in  $R_g$ .

For (iii), a map from  $R[x]/(xg - 1) \rightarrow S$  is the same as  $x \mapsto s$  where  $sg = 1$ .

For (iv), if  $g$  is a unit in  $R$ , then  $R[x]/(xg - 1)$  can be written as  $R[x]/(x - g^{-1}) \cong R$ .  $\square$

As an example,  $\mathbb{Z}_0 = 0$ . This is because  $\mathbb{Z}[x]/(x(0) - 1) = \mathbb{Z}[x]/(1) = 0$ . More generally, if  $g$  is nilpotent, i.e.,  $g^N = 0$ , then  $R_g = 0$ , since  $g^N r = 0$ . **I kinda got lost at these examples cuz the board was too dusty.**

**Theorem 5.5.**  $|\text{Spec } R_g| \cong D(g) \subset |\text{Spec } R|$ .

*Proof.* Consider the map  $\phi: R \rightarrow R_g$ , so  $\phi^*: |\text{Spec } R_g| \rightarrow |\text{Spec } R|$ . First, we can show that  $\text{Im } \phi^* \subset D(g)$ . Indeed,  $(\phi^*)^{-1}(D(g)) = Z(\phi(g))$  (since zero sets pullback the way you expect), which is equal to  $\phi$ .

Now, we want to show that  $\phi^*$  is injective, i.e., we can recover  $\mathfrak{p}$  from  $\phi^* \mathfrak{p} = \phi^{-1}(\mathfrak{p})$ . We claim that  $(\phi(\phi^{-1}(\mathfrak{p}))) = \mathfrak{p}$ . The non-trivial containment is that  $\mathfrak{p} \subset \phi(\phi^{-1}(\mathfrak{p}))$ . Given  $r/g$  in  $\mathfrak{p}$ , we have  $r \in \mathfrak{p}$ , so  $r \in \phi^{-1}(\mathfrak{p})$ , so  $r \in \phi(\phi^{-1}(\mathfrak{p}))$ , whence  $r/g \in (\phi(\phi^{-1}(\mathfrak{p})))$ .

It remains to check that  $\phi^*$  is surjective. Take  $\mathfrak{q} \in D(g) \subset \text{Spec } R$ , and note that  $R \rightarrow \kappa(\mathfrak{q})$  sends  $g$  to a unit. Thus, we can factor the map as  $R \rightarrow R_g \rightarrow \kappa(\mathfrak{q})$ . Taking the spectrum of this factorization, we get  $\text{Spec } R \leftarrow \text{Spec } R_g \leftarrow \text{Spec } \kappa(\mathfrak{q})$ , so  $\text{Spec } R_g$  contains  $\mathfrak{q}$ .

Finally, given  $D(h) \in \text{Spec } R_g$ , we want to show that  $D(h)$  is open in  $\text{Spec } R_g$ . We have that  $h = f/g^i$  for some  $i$  and  $f \in R$ ; for  $\phi: R \rightarrow R_g$ , without loss of generality,  $i = 0$ , so  $h = \phi(f)$  in  $R_g$ . Thus,

$$(\phi^*)^{-1}D(f) = D(\phi(f)) = D(h);$$

but  $(\phi^*)^{-1}$  is just restriction, so  $D(h) = D(f)$  is open in  $\text{Spec } R$ .  $\square$

He will correct this proof next class, apparently. **“iunno. fix this proof. it’s an exercise”**

## Day 6: Localization and Germs (Sep. 24, 2026)

Last class, we saw that, for  $\phi: R \rightarrow R_g$ , we have that  $\phi^*: \text{Spec } R_g \hookrightarrow \text{Spec } R$  has image  $D(g) \subset \text{Spec } R$ . We want to show that it is homeomorphism onto its image. Given an open set  $D(f/g^i) \subset \text{Spec } R_g$ , we want an open set  $U$  in  $\text{Spec } R$  such that  $(\phi^*)^{-1}U = D(f/g^i)$ . Indeed,  $D(f/g^i)_{\text{Spec } R_g} = D(f)$ , so we may take  $U = D(f) \subset \text{Spec } R$ .

“By now, we’re getting *really* close to the definition of a scheme... next week, we’ll be ‘full scheme ahead’.”

**Definition 6.1.** An ideal  $I \subset R$  is said to be *radical* if  $I = \sqrt{I} := \{r \in R \mid r^n \in I \text{ for some } n\}$ .  $\sqrt{I}$  is called the *radical* of  $I$ .

Note that the radical of zero,  $\sqrt{(0)}$ , is equal to the set of all nilpotent elements of  $R$ , which we see by unraveling the definitions. Recall that  $f(\mathfrak{p}) = 0$  for all  $\mathfrak{p} \in \text{Spec } R$  if  $f$  is nilpotent. Moreover,  $I$  being radical is equivalent to  $R/I$  being reduced, i.e., it does not have any nilpotent elements. In general,  $(R/I)_{\text{red}} := (R/I)/\sqrt{(0)} = R/\sqrt{I}$ .

**Theorem 6.2.** Let  $I \subset R$  be an ideal; then

$$\sqrt{I} = \bigcap_{\mathfrak{p} \supset I} \mathfrak{p}.$$

In particular,  $\sqrt{(0)}$  is the intersection of all prime ideals  $\mathfrak{p} \in \text{Spec } R$ , i.e.,  $f(\mathfrak{p}) = 0$  for all  $\mathfrak{p}$  if and only if  $f$  is nilpotent.

*Proof.* By quotienting by  $I$ , we can reduce the question to the case  $I = (0)$ . We know

$$\sqrt{(0)} \subset \bigcap_{\mathfrak{p} \in \text{Spec } R} \mathfrak{p}$$

because if  $r$  is nilpotent, then  $r^n = 0 \in \mathfrak{p}$  implies  $r \in \mathfrak{p}$ . Conversely, we want to show that if  $f \in \bigcap_{\mathfrak{p} \in \text{Spec } R} \mathfrak{p}$ , then  $f$  is nilpotent. Indeed,  $D(f) = \emptyset$ , i.e., there is no ideal in which  $f$  does not vanish in. Since  $D(f) = \text{Spec } R_f$ , we see that this means  $R_f$  is the zero ring, i.e., in  $R_f$ , we have  $1 = 0$ . Thus, in  $R$ ,  $f^n \cdot 1 = 0$  for some  $n$ , and so  $f^n = 0$ .  $\square$

Note that we implicitly use Zorn’s lemma above, i.e., if  $\text{Spec } R_f = \emptyset$ , then  $R_f$  is the zero ring.

**Theorem 6.3.**  $Z(I) = Z(\sqrt{I})$ , and  $Z$  gives a contravariant equivalence.

$$\{\text{radical ideals in } R\} \rightarrow \{\text{closed subsets of } \text{Spec } R\}.$$

*Proof.* For the first part,  $Z(I) = Z(\bigcap_{\mathfrak{p} \supset I} \mathfrak{p}) = Z(\sqrt{I})$ . For the second part, we ask, how do we recover the radical  $I$  from the zero locus? Since  $Z(I) = \{\mathfrak{p} \mid \mathfrak{p} \supset I\}$ , we see that

$$\bigcap_{\mathfrak{p} \in Z(I)} \mathfrak{p} = \sqrt{I} = I. \quad \square$$

Given a closed set  $V$ , we may define  $I(V) = \bigcap_{\mathfrak{p} \in V} \mathfrak{p}$ , and we have  $Z(I(V)) = V$  and  $I(Z(J)) = \sqrt{J}$ .

**Corollary 6.4.** We have the following,

- (i)  $D(f) = \emptyset$  if and only if  $Z(f) = \text{Spec } R$  if and only if  $f$  is nilpotent.
- (ii)  $D(f) = \text{Spec } R$  if and only if  $Z(f) = \emptyset$  if and only if  $f$  is a unit.
- (iii)  $\text{Spec } R = \bigcup_{\alpha \in \mathcal{A}} D(f_\alpha)$  if and only if  $Z(\{f_\alpha \mid \alpha \in \mathcal{A}\}) = \emptyset$  if and only if  $\{f_\alpha \mid \alpha \in \mathcal{A}\} = (1)$ ,

As a throwback to day 1 of class, observe that  $x^2 + 1$  does not vanish on any real number, but it is also not invertible. We may look at the prime ideal  $\mathfrak{p} = (x^2 + 1)$  in  $\mathbb{R}[x]$  to see  $x^2 + 1 = 0$  in  $\kappa(\mathfrak{p})$ , so (ii) does not apply. Also note that, in classical algebraic geometry, the Nullstellensatz states that if  $k = \bar{k}$ , and  $f_1, \dots, f_r$  have no common solution, then there exists  $g_1, \dots, g_r$  such that  $\sum_i g_i f_i = 1$ .

*Proof.* We just proved (i) earlier. For (ii), we have that  $I(Z(f)) = \sqrt{(f)}$ , so  $I(\emptyset) = (1)$ , then  $1^n \in (f)$  for some  $n$ , which tells us  $1 \in (f)$ , whence  $f$  is a unit. For (iii),

$$I(Z(\{f_\alpha \mid \alpha \in \mathcal{A}\})) = \sqrt{(f_\alpha \mid \alpha \in \mathcal{A})},$$

and the left hand side is equal to  $I(\emptyset) = (1)$ , so  $1^n \in (f_\alpha \mid \alpha \in \mathcal{A})$ , whence  $(f_\alpha \mid \alpha \in \mathcal{A}) = (1)$ .  $\square$

**Corollary 6.5.**  $\text{Spec } R$  is quasi-compact.

*Proof.* By (iii) from 6.4, we have that  $(f_\alpha \mid \alpha \in \mathcal{A}) = (1)$ , so there exists  $\sum_{i=1}^n f_i g_i = 1$ , which implies that

$$\text{Spec } R = \bigcup_{i=1}^n D(f_i). \quad \square$$

In this manner, we may think  $\text{Spec } R_p \stackrel{“=”}{=} D(f) \subset \text{Spec } R$ . The problem with writing equality is that, if we had another function  $g$  such that  $D(f) = D(g)$ , then we might want to have a different canonical ring to describe  $D(g)$ . For example,  $D(x) = D(x^2)$ , and  $\text{Spec } \mathbb{C}[x]_x = Z(xy - 1)$ , and  $\text{Spec } \mathbb{C}[x]_{x^2} = Z(x^2y - 1)$ .

**Theorem 6.6.** If  $D(f) = D(g)$ , then there exists a unique  $R$ -algebra map between  $R_f$  and  $R_g$ . In particular, this map is an isomorphism.

*Proof.* Look at  $Z(f)_{\text{Spec } R_g} = \emptyset$ . Then  $f$  is a unit in  $R_g$ , so there is a canonical map  $R_f \rightarrow R_g$ , and vice versa.  $\square$

**Corollary 6.7.** For a basic open set  $U$ , the “ring of functions” is  $R$  localized at all  $f \in R$  such that  $f$  does not vanish on  $U$ .

**Theorem 6.8.** The localization of a localization is the localization of the product, i.e.,  $(R_f)_g = R_{fg}$ . Another way to think about this is to say that restricting from  $D(f)$  to  $D(g)_{D(f)}$  is the same as directly restricting to  $D(fg)$ .

*Proof.* The proof is left as a homework exercise.  $\square$

We now talk about germs of functions. A germ is a function defined on a neighborhood of a point; specifically, we define a *germ* at  $\mathfrak{p} \in \text{Spec } R$  to be a pair  $(f, U)$  (where  $U \ni \mathfrak{p}$  is a basic open set) such that  $f \in R_g$  up to an equivalence relation that  $(f_1, U_1) \sim (f_2, U_2)$  provided there exists a basic open set  $U_3 \subset U_1 \cap U_2$  such that  $\mathfrak{p} \in U_3$  and  $(f_1|_{U_3}, U_3) = (f_2|_{U_3}, U_3)$ . We may regard  $f_1|_{U_3}$  as the inverse of  $f$  in  $\mathcal{O}(U_3)$ .

He then gave an example with parabolas on the board.

**Definition 6.9.** The localization of  $R$  at  $\mathfrak{p}$  is given by  $R_{\mathfrak{p}} := R[\{x_f \mid f \notin \mathfrak{p}\}]/\{1 - x_f f \mid f \notin \mathfrak{p}\}$ .

**Theorem 6.10.**  $R_p$  is isomorphic to the ring of germs of  $R$  at  $p$ .

*Proof.* We can map  $R_p$  to the ring of germs by  $r \mapsto (r, \text{Spec } R)$  and  $x_f \mapsto (f^{-1}, D(f))$ . Clearly, this is well-defined based on how we wrote the definitions earlier. Surjectivity is also obvious. What about injectivity, though? Suppose we have some element of  $R_p$ . This only involves finitely many of the formal inverses, so we can collect denominators, so we may write  $r/g$ . Suppose  $r/g \mapsto 0$ , i.e.,  $(r/g, D(g)) = 0$ . This means, there exists  $h$  such that  $D(h) \subset D(g)$ , so that  $d/g|_{D(h)} = 0$ . Equivalently,  $r|_{D(h)} = 0$ , so  $h^N r = 0$  in  $R$ , so  $r = 0$  in  $R_p$ , as this is a unit in  $R$ , so  $r/g = 0$  in  $R$ .  $\square$